Linear Algebra Mastery

Vectors and Vector Operations

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1 Introduction

A vector is an ordered list of numbers with both magnitude and direction. Vectors are very common in machine learning applications. Whenever a linear regression model is trained, the predictions are presented using vector-linear combination method e.g. $\hat{y} = w_1x_1 + w_2x_2 + \cdots + w_nx_n$. There are many other applications of vectors in real world, they are not just limited to linear regression modelling. Some of them includes:

1. Scaling a feature, done in feature normalization
2. Understanding span of all possible predictions a model can make.
3. Compute model weights, from $A\vec{x} = b$, given that $\vec{x}$ is a vector with the model weights.

2 Formal Definition

2.1 Vector addition

$$\vec{u} + \vec{v} = \begin{pmatrix} u_1 + v_1 \\ u_2 + v_2 \\ \vdots \\ u_n + v_n \end{pmatrix} \qquad \text{Given that: } \vec{u}, \vec{v} \in \mathbb{R}^n$$

Note: Vector addition is *commutative* meaning $\vec{u} + \vec{v} = \vec{v} + \vec{u}$

2.2 Scalar multiplication

$$c\vec{v} = \begin{pmatrix} cv_1 \\ cv_2 \\ \vdots \\ cv_n \end{pmatrix} \qquad \text{Given that: } c \in \mathbb{R}, \quad \vec{v} \in \mathbb{R}^n$$

Scalar multiplication meaning:

1. If $c > 1$: Positive stretching
2. If $0 < c < 1$: Diminishing/shrinking vector

3. If $c < 0$: Flipping
4. If $c = 0$: Collapsing to a zero vector

2.3 Linear combination

Scaling vectors and adding them together.

$$c_1\vec{v}_1 + c_2\vec{v}_2 + \cdots + c_n\vec{v}_n$$
$$\text{Where: } c_i \in \mathbb{R} \text{ and } \vec{v}_i \in \mathbb{R}^n$$

2.4 Span

Span is a set of all linear combinations of a set of vectors.

$$span\{\vec{v}_1, \vec{v}_2\} = span\{c_1\vec{v}_1 + c_2v_2 \mid c_i \in \mathbb{R}, \quad \vec{v}_i \in \mathbb{R}^n\}$$

Geometry explanation:

1. Span of a non zero vector is a line through the origin.
2. Span of a two non-parallel vectors, spans the entire $\mathbb{R}^2$ space.
3. Span of two parallel vectors is a line in $\mathbb{R}^2$

3 Python Results

Running the script with $\vec{u} = (1, 3)$ and $\vec{v} = (2, -1)$ produced $\vec{v}+\vec{u} = (3, 2)$, confirming commutative, $\vec{v}+\vec{u} = (3, 2)$. The rank check confirmed that $\vec{v}_1 = (1, 0)$ and $\vec{v}_2 = (0, 1)$ spans $\mathbb{R}^2$. The two vectors are not parallel, so they provide new information and spans the entire plane. $\vec{v}_3 = (2, 4)$ and $\vec{v}_4 = (1, 2)$ had a rank $= 1$, meaning that the two vectors are parallel, therefore the two vectors form a line in $\mathbb{R}^2$.

4 ML Connection

1. Adding features in a feature matrix
2. Normalizing features
3. Presenting predictions from a linear regression model. Linear regression predictions is a linear combination of weights and feature vector e.g. $\hat{y} = w_1x_1 + w_2x_3 + \cdots + w_nx_n$
4. Span of a feature matrix should be the same as the one for target vector. If the target vector lies outside the span, no exact solution exists.

5 Reflection

Todays lesson reinforced my understanding of vectors as a fundamental building block for linear algebra and machine learning. The concept I found challenging today was the formal notation of span. Specifically distinguishing between set label $span\{\vec{v}_1, \vec{v}_2\}$ and linear combinations $span\{c_1\vec{v}_1 + \cdot + c_n\vec{v}_n\}$. What fascinate me the most, is how linear regression model predictions are presented using linear combination of weights and features.